UNIT – II

LINEAR PROGRAMMING

1.	Define linear programming problem (LPP) with an example.
2.	Define optimization of LPP.
3.	Define objective function in linear programming problem.
4.	Define decision variables in linear programming problem.
5.	Define constraints in linear programming problem.
6.	Explain the non-negativity constraints in LPP.
7.	Explain the formation of linear programming problem.
8.	Form the linear programming problem to maximize profit a manufacturer produces two of models M_1 and M_2. Each M_1 model requires 4 hours of grinding and 2 hours of pol whereas each M_2 model requires 2 hours of grinding and 5 hours of polishing manufacturer has 2 grinders and 3 polishers. Each grinder works for 40 hours a week and polisher works for 60 hours a week. Profit on an M_1 model is Rs.3 and on as M_2 model is Whatever is produced in a week is sold in the market. How should the manufacture alloca production capacity to the two types of models so that he may make the maximum prof week.
9.	A manufacturer produces nuts and bolts for industrial machinery. It takes 1 hour of wo machine A and 3 hours on machine B to produce a package of nuts, while it takes 3 hou machines A and 1hours on machine B to produce a package of bolts. He earns a pro Rs.2.50 per package on nuts and Rs.1 per package on bolts. Form a linear program problem to maximize his profit, if he operates each machine for at the most 12 hours a day
10.	A person wants to invest an amount up to Rs.30,00. His well-wisher recommends investin two types of bonds A and B. Bond A yielding 12% return on the amount invested and bon yielding 18% return on the amount invested. After some consideration, he decides to inves least Rs.8000 in bond A and not more than Rs. 15,000 in bond B. He also wants to inves least as much in bond A as in bond B. What should his well-wisher suggest if he wants maximize his return on investments. Formulate LPP.
11.	Define solution of the LPP.

12.	Define feasible solution of the LPP.
13.	Define an optimal solution of the LPP.
14.	Define feasible region of the LPP.
15.	Define convex region of the LPP with an example.
16.	Explain the working procedure of the graphical method to solve given LPP.
17.	Find the maximum value of $Z = 3x_1 + 4x_2$ subject to the constraints $4x_1 + 2x_2 \le 80$, $4x_1 + 2x_2 \le 80$, $2x_1 + 5x_2 \le 180$, $x_1, x_2 \ge 0$, by solving LPP using graphical method.
18.	Find the maximum value of $Z = 2x + 3y$ subject to the constraints $x + y \le 30$, $y \ge 3$, $x - y \ge 0$, $0 \le x \le 20$ and $0 \le y \le 12$ using graphical method.
19.	Maximize $z = x + 1.5y$ subject to the constraints $x + 2y \le 160$, $3x + 2y \le 240$, given $x, y \ge 0$ by graphical method.
20.	Minimize $z = 5x + 4y$ subject to the constraints $x + 2y \le 10$, $x + y \ge 8$, $2x + y \ge 12$, $x \ge 0$, $y \ge 0$ by graphical method.
21.	Use the graphical method to Minimize $z = 20x_1 + 10x_2$ subject to the constraints $x_1 + 2x_2 \le 40$, $3x_1 + x_2 \ge 30$, $4x_1 + 3x_2 \ge 60$, $x_1 \ge 0$, $x_2 \ge 0$.
22.	Minimize $z = 30x + 20y$ subject to the constraints $x - y \le 1$, $x + y \ge 3$, $y \le 4$ $x, y \ge 0$ by graphical method.
23.	Using the graphical method to solve the following LPP, Minimize $z = 8x + 5y$ subject to the constraints $6x + y \ge 21$, $x + 3y \ge 12$, $x + y \le 10$ and $x_1 \ge 0$, $x_2 \ge 0$.
24.	A company manufactures two types of cloths, using three different colours of wool. One-yard length of type A cloth requires 4 Oz of red wool, 5 Oz of green wool and 3 Oz of yellow wool. One-yard length of type B cloth requires 5 Oz of red wool, 2 Oz of green wool and 8 Oz of yellow wool. The wool available for manufacturer is 1000 Oz of red wool, 1000 Oz of green wool and 1200 Oz of yellow wool. The manufacturer can make a profit of Rs.5 on one yard of type A cloth and Rs.3 on one yard of type B cloth. Find the best combination of the quantities of type A cloth and type B cloth which gives him maximum profit by solving the LPP graphically.
25.	Show that the following LPP does not have any feasible solution. Objective function for maximization $z = 20x + 30y$ subject to the constraints $3x + 4y \le 24$, $7x + 9y \ge 63$ and $x \ge 0$, $y \ge 0$ by applying graphical method.
26.	A factory uses 3 types of machines to produce two types of electronic gadgets. The first gadget requites i_1 hours 12, 4 and 2 respectively on the three types of machines. The second gadget requires i_2 hours 6, 10 and 3 on the machines respectively. The total available time in hours

	respectively on the machines 6000, 4000, 1800. If the two types of gadgets respective[ly] a profit of rupees 400 and 1000 find the number of gadgets of each type to be produ[ced] getting the maximum profit by applying graphical method.
27.	Explain general LPP with an example.
28.	Explain the canonical form of LPP with an example.
29.	Explain the standard form of LPP with an example.
30.	Explain the working procedure to convert any LPP to standard for of LPP.
31.	Convert the LPP, Maximize $Z = 3x_1 + 5x_2 + 7x_3$, subject to the constraints $6x_1 - 4x$ $3x_1 + 2x_2 + 5x_3 \geq 11$, $4x_1 + 3x_3 \leq 2$, $x_1 \geq 0$, $x_2 \geq 0$ to the standard form of the LPP.
32.	Express the LPP, Minimize $Z = 3x_1 + 4x_2$, subject to $2x_1 - x_2 - 3x_3 = -4$, $3x_1 + 5x_2 + x_4 = 1$ $x_1 - 2x_2 = 12$, $x_1 \geq 0$, $x_3 \geq 0$, $x_4 \geq 0$ to the standard form of the LPP.
33.	Define slack variables of the LPP.
34.	Define surplus variables of the LPP.
35.	Define artificial variables of the LPP.
36.	Define the basic variables and non-basic variables in LPP.
37.	Define basic solution of the LPP.
38.	Define degenerate solution and non-degenerate solution of the LPP.
39.	Define incoming variables and outgoing variables in LPP.
40.	Define optimal basic feasible solution or optimal solution of the LPP.
41.	Explain working procedure of simplex method to solve the given LPP.
42.	Explain working procedure of big M simplex method to solve the given LPP.
43.	Find all the basic solutions of the system of equations $2x_1 + x_2 + 4x_3 = 11$, $3x_1 + x_2 + 5x_3$,identifying in each case the basic and non-basic variables. Investigate whether the b solutions are degenerate basic solution or not. Hence find the basic feasible solution of system.
44.	Find an optimal solution to the LPP $2x_1 + 3x_2 - x_3 + 4x_4 = 8$, $x_1 - 2x_2 + 6x_3 - 7x_4 =$ $x_1, x_2, x_3, x_4 \geq 0$, Maximize $Z = 2x_1 + 3x_2 + 4x_3 + 7x_4$ by computing all basic solutions and t finding one that maximizes the objective function.
45.	Use simplex method solve the LPP to maximize $z = x + 1.5y$ subject to the constrai

	$x+2y \le 160$, $3x+2y \le 240$, given $x, y \ge 0$.
46.	Solve the LPP to maximize $P = 2x+3y+z$ subject to the constraints $x+3y+2z \le 11$, $x+2y+5z \le 19$, $3x+y+4z \le 25$ given $x, \ge 0$, $y \ge 0$, $z \ge 0$ by applying simplex method.
47.	Using simple method solve the LPP to maximize $z = 5x_1 + 3x_2$ subject to the constraints $x_1 + x_2 \le 2$, $5x_1 + 2x_2 \le 10$, $3x_1 + 8x_2 \le 12$, $x_1 \ge 0$, $x_2 \ge 0$.
48.	Use simple method solve the LPP to maximize $z = 3x_1 + 4x_2$ subject to the constraints $2x_1 + x_2 \le 40$, $2x_1 + 5x_2 \le 180$, $x_1 \ge 0$, $x_2 \ge 0$.
49.	Use simple method solve the LPP to maximize $z = 2x + 4y$ subject to the constraints $3x + y \le 22$, $2x + 3y \le 24$, $x \ge 0$, $y \ge 0$.
50.	Use simple method solve the LPP to minimize $P = x - 3y + 2z$ subject to the constraints $3x - y + 2z \le 7$, $-2x + 4y \le 12$, $-4x + 3y + 8z \le 10$, $x \ge 0$, $y \ge 0$, $z \ge 0$.
51.	Using Big – M method solve the LPP to minimize $z = 2x_1 + x_2$ subject to $3x_1 + x_2 = 3$, $4x_1 + 3x_2 \ge 6$, $x_1 + 2x_2 \le 3$, $x_1 \ge 0$, $x_2 \ge 0$.
52.	Solve the LPP to maximize $z = 3x_1 + 2x_2$ subject to the constraints $2x_1 + x_2 \le 2$, $3x_1 + 4x_2 \ge 12$, $x_1 \ge 0$, $x_2 \ge 0$ by using Big-M method.
53.	Explain two phase simplex method in LPP.
54.	Explain formation of duality in linear programming problem.
55.	Write the dual of the following LPP to minimize $z = 3x_1 - 2x_2 + 4x_3$ subject to $3x_1 + 5x_2 + 4x_3 \ge 7$, $6x_1 + x_2 + 3x_3 \ge 4$, $7x_1 - 2x_2 - x_3 \le 10$, $x_1 - 2x + 5x_3 \ge 3$, $4x_1 + 7x_2 - 2x_3 \ge 2$ $x_1 \ge 0$, $x_2 \ge 0$, $x_3 \ge 0$.
56.	Construct the dual of the LPP to maximize $z = 4x_1 + 9x_2 + 2x_3$ subject to $2x_1 + 3x_2 + 2x_3 \le 7$, $3x - 2x_2 4x_3 = 5$, $x_1 \ge 0$, $x_2 \ge 0$, $x_3 \ge 0$.
57.	Use simplex method solve the LPP to maximize $z = x + 1.5y$ subject to the constraints $x + 2y \le 160$, $3x + 2y \le 240$, given $x, y \ge 0$.